

R. Deepasri (192511309)

EX .NO: 28 TO 37 (Google Cloud)

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 **Google Cloud** Select a project Search (/) for resources, docs, products, and more

 CLOUD SHELL
Terminal cloudshell x +

```
elcome to Cloud Shell! Type "help" to get started.
o set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
ou can view your projects by running `gcloud projects list`.
eepasrir7@cloudshell:~$
```

 CLOUD SHELL
Terminal cloudshell x +

```
GNU nano 7.2
print("Hello from Google Cloud!")
print("Welcome to Cloud Computing")
```

 Start your Free Trial with \$300 in credit. Don't worry—

 **Google Cloud** Select a project

 CLOUD SHELL
Terminal cloudshell x +

```
GNU nano 7.2
print("Hello from Google Cloud")
print("Welcome to Cloud Computing")
```



```
Welcome to Cloud Shell! Type "help" to get started.
To set your Cloud Platform project in this session use `gcloud config set project [PROJECT_ID]`.
You can view your projects by running `gcloud projects list`.
deepasrir7@cloudshell:~$ nano hello.py
deepasrir7@cloudshell:~$ python3 hello.py
Hello from Google Cloud!
Welcome to Cloud Computing
deepasrir7@cloudshell:~$
```

```
Welcome to Cloud Computing
deepasrir7@cloudshell:~$ mkdir cloudlab
deepasrir7@cloudshell:~$ cd cloudlab
deepasrir7@cloudshell:~/cloudlab$ echo "Welcome to Google Cloud" > sample.txt
deepasrir7@cloudshell:~/cloudlab$ cat sample.txt
Welcome to Google Cloud
deepasrir7@cloudshell:~/cloudlab$
```

```
deepasrir7@cloudshell:~/cloudlab$ cat sample.txt
Welcome to Google Cloud
deepasrir7@cloudshell:~/cloudlab$ ls
sample.txt
deepasrir7@cloudshell:~/cloudlab$ cp sample.txt backup.txt
deepasrir7@cloudshell:~/cloudlab$ ls
backup.txt  sample.txt
deepasrir7@cloudshell:~/cloudlab$
```

 Google Cloud Select a project

 CLOUD SHELL
Terminal cloudshell x + ▼

GNU nano 7.2

```
<html>
<body>
<h1>Google Cloud Practical</h1>
<p>My first cloud web application</p>
</body>
</html>
```

 8080-cs-0e5c2bd9-0543-4e01-a1e8-c419ac1f30

Google Cloud Practical

My first cloud web application

```

deepasrir7@cloudshell:~$ mkdir cloudlab
mkdir: cannot create directory 'cloudlab': File exists
deepasrir7@cloudshell:~$ cd cloudlab
deepasrir7@cloudshell:~/cloudlab$ echo "Welcome to Google Cloud"> sample.txt
deepasrir7@cloudshell:~/cloudlab$ cat sample.txt
Welcome to Google Cloud
deepasrir7@cloudshell:~/cloudlab$ ls
backup.txt  index.html  sample.txt
deepasrir7@cloudshell:~/cloudlab$ cp sample.txt backup.txt
deepasrir7@cloudshell:~/cloudlab$ ls
backup.txt  index.html  sample.txt
deepasrir7@cloudshell:~/cloudlab$ nano index.html
deepasrir7@cloudshell:~/cloudlab$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
127.0.0.1 - - [28/Aug/2026 07:59:28] "GET /?authuser=0 HTTP/1.1" 200 -
127.0.0.1 - - [28/Aug/2026 07:59:28] code 404, message File not found
127.0.0.1 - - [28/Aug/2026 07:59:28] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [28/Aug/2026 08:00:44] "GET /?authuser=0 HTTP/1.1" 304 -

```

terminal cloudshell x + ▾

```

deepasrir7@cloudshell:~/cloudlab$ nano index.html
deepasrir7@cloudshell:~/cloudlab$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
127.0.0.1 - - [28/Aug/2026 07:59:28] "GET /?authuser=0 HTTP/1.1" 200 -
127.0.0.1 - - [28/Aug/2026 07:59:28] code 404, message File not found
127.0.0.1 - - [28/Aug/2026 07:59:28] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [28/Aug/2026 08:00:44] "GET /?authuser=0 HTTP/1.1" 304 -
^C
Keyboard interrupt received, exiting.
deepasrir7@cloudshell:~/cloudlab$ ^C
deepasrir7@cloudshell:~/cloudlab$ free -h
              total        used        free      shared  buff/cache   available
Mem:           7.8Gi        616Mi        6.1Gi         1.1Mi         1.2Gi         7.2Gi
Swap:           0B           0B           0B

deepasrir7@cloudshell:~/cloudlab$ df -h
Filesystem      Size  Used Avail Use% Mounted on
overlay          119G   65G   54G   55% /
tmpfs            64M    0    64M    0% /dev
/dev/sdal        119G   65G   54G   55% /root
/dev/disk/by-id/google-home-part1 4.8G   3.2M   4.6G    1% /home
shm              64M    0    64M    0% /dev/shm

deepasrir7@cloudshell:~/cloudlab$ lscpu
-bash: lscpu: command not found
deepasrir7@cloudshell:~/cloudlab$ iscpu
-bash: iscpu: command not found
deepasrir7@cloudshell:~/cloudlab$ lscpu
Architecture:        x86_64
CPU op-mode(s):      32-bit, 64-bit
Address sizes:        46 bits physical, 48 bits virtual
Byte Order:           Little Endian
CPU(s):               2
On-line CPU(s) list: 0,1
Vendor ID:            GenuineIntel
Model name:           Intel(R) Xeon(R) CPU @ 2.20GHz

```



CLOUD SHELL

Terminal

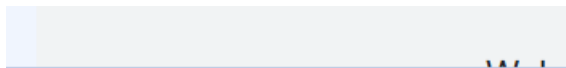
cloudshell x + ▾

```
Vendor ID: GenuineIntel
Model name: Intel(R) Xeon(R) CPU @ 2.20GHz
CPU family: 6
Model: 79
Thread(s) per core: 2
Core(s) per socket: 1
Socket(s): 1
Stepping: 0
top - 08:06:45 up 32 min,  0 user,  load average: 0.11, 0.05, 0.02
Tasks: 24 total,  1 running, 23 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.3 us,  0.3 sy,  0.0 ni, 99.3 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7947.4 total, 6300.0 free,  606.2 used, 1280.1 buff/cache
MiB Swap:  0.0 total,  0.0 free,  0.0 used. 7341.1 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
25	root	20	0	38796	28072	10840	S	0.3	0.3	0:00.66	python
1	root	20	0	4324	3356	3072	S	0.0	0.0	0:00.07	bash
9	syslog	20	0	222140	3848	2500	S	0.0	0.0	0:00.01	rsyslogd
26	root	20	0	6196	2620	2380	S	0.0	0.0	0:00.00	logger
74	root	10	-10	12020	4072	2956	S	0.0	0.1	0:00.01	sshd
78	root	10	-10	12016	2876	1768	S	0.0	0.0	0:00.00	sshd
328	root	20	0	2025108	88252	64336	S	0.0	1.1	0:00.71	dockerd
426	root	20	0	1267452	8492	7140	S	0.0	0.1	0:00.02	editor-proxy
427	root	20	0	16328	6412	5488	S	0.0	0.1	0:00.01	sudo
447	root	20	0	1227876	2968	2164	S	0.0	0.0	0:00.00	tmux-agent
468	root	20	0	1747472	42524	30092	S	0.0	0.5	0:01.33	containerd
654	root	20	0	2696	1540	1440	S	0.0	0.0	0:00.00	sleep
690	root	10	-10	13992	9752	8376	S	0.0	0.1	0:00.01	sshd
691	root	10	-10	13992	9760	8384	S	0.0	0.1	0:00.02	sshd
694	deepasr+	10	-10	14252	6448	4792	S	0.0	0.1	0:00.08	sshd
695	deepasr+	10	-10	7340	3792	3544	S	0.0	0.0	0:00.00	bash
696	deepasr+	10	-10	7340	3644	3384	S	0.0	0.0	0:00.00	start-shell.sh
697	deepasr+	10	-10	6876	3516	3216	S	0.0	0.0	0:00.00	tmux
699	deepasr+	10	-10	7276	3036	2256	S	0.0	0.0	0:00.10	tmux
700	deepasr+	10	-10	9312	6316	3932	S	0.0	0.1	0:00.10	bash

```
top - 08:07:18 up 33 min,  0 user,  load average: 0.01/ 0.01/ 0.01
Tasks: 24 total,  1 running, 23 sleeping,  0 stopped,  0 zombie
%Cpu(s):  2.0 us,  1.0 sy,  0.0 ni, 96.8 id,  0.0 wa,  0.0 hi,  0.2 si,  0.0 st
MiB Mem : 7947.4 total, 6295.1 free,  610.9 used, 1280.3 buff/cache
MiB Swap:  0.0 total,  0.0 free,  0.0 used. 7336.4 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	4324	3356	3072	S	0.0	0.0	0:00.07	bash
9	syslog	20	0	222140	3848	2500	S	0.0	0.0	0:00.01	rsyslogd
25	root	20	0	38796	28072	10840	S	0.0	0.3	0:00.66	python
26	root	20	0	6196	2620	2380	S	0.0	0.0	0:00.00	logger
74	root	10	-10	12020	4072	2956	S	0.0	0.1	0:00.01	sshd
78	root	10	-10	12016	2876	1768	S	0.0	0.0	0:00.00	sshd
328	root	20	0	2025108	88252	64336	S	0.0	1.1	0:00.72	dockerd
426	root	20	0	1267452	8492	7140	S	0.0	0.1	0:00.02	editor-proxy
427	root	20	0	16328	6412	5488	S	0.0	0.1	0:00.01	sudo
447	root	20	0	1227876	2968	2164	S	0.0	0.0	0:00.00	tmux-agent
468	root	20	0	1747472	42440	30092	S	0.0	0.5	0:01.41	containerd
654	root	20	0	2696	1540	1440	S	0.0	0.0	0:00.00	sleep
690	root	10	-10	13992	9752	8376	S	0.0	0.1	0:00.01	sshd
691	root	10	-10	13992	9760	8384	S	0.0	0.1	0:00.02	sshd
694	deepasr+	10	-10	14252	6448	4792	S	0.0	0.1	0:00.09	sshd
695	deepasr+	10	-10	7340	3792	3544	S	0.0	0.0	0:00.00	bash
696	deepasr+	10	-10	7340	3644	3384	S	0.0	0.0	0:00.00	start-shell.sh
697	deepasr+	10	-10	6876	3516	3216	S	0.0	0.0	0:00.00	tmux
699	deepasr+	10	-10	7276	3036	2256	S	0.0	0.0	0:00.12	tmux
700	deepasr+	10	-10	9312	6316	3932	S	0.0	0.1	0:00.10	bash
896	deepasr+	10	-10	14252	6364	4708	S	0.0	0.1	0:00.00	sshd
989	deepasr+	10	-10	7340	3736	3484	S	0.0	0.0	0:00.00	bash
997	deepasr+	10	-10	7604	4488	3888	S	0.0	0.1	0:00.00	bash
1148	deepasr+	10	-10	11840	5964	3796	R	0.0	0.1	0:00.06	top



CLOUD SHELL

Terminal

cloudshell x +

GNU nano 7.2

```
1 #include<stdio.h>
2 int main()
3 {
4     int a=10,b=20;
5     printf("sum=%d\n",a+b);
6     return 0;
7 }
8
9
```

426	root	20	0	1267432	8432	7140	S	0.0	0
427	root	20	0	16328	6412	5488	S	0.0	0
447	root	20	0	1227876	2968	2164	S	0.0	0
468	root	20	0	1747472	42440	30092	S	0.0	0
654	root	20	0	2696	1540	1440	S	0.0	0
690	root	10	-10	13992	9752	8376	S	0.0	0
691	root	10	-10	13992	9760	8384	S	0.0	0
694	deepasr+	10	-10	14252	6448	4792	S	0.0	0
695	deepasr+	10	-10	7340	3792	3544	S	0.0	0
696	deepasr+	10	-10	7340	3644	3384	S	0.0	0
697	deepasr+	10	-10	6876	3516	3216	S	0.0	0
699	deepasr+	10	-10	7276	3036	2256	S	0.0	0
700	deepasr+	10	-10	9312	6316	3932	S	0.0	0
896	deepasr+	10	-10	14252	6364	4708	S	0.0	0
989	deepasr+	10	-10	7340	3736	3484	S	0.0	0
997	deepasr+	10	-10	7604	4488	3888	S	0.0	0
1148	deepasr+	10	-10	11840	5964	3796	R	0.0	0

```
deepasrir7@cloudshell:~/cloudlab$ nano add.c
deepasrir7@cloudshell:~/cloudlab$ gcc add.c -o add
deepasrir7@cloudshell:~/cloudlab$ ./add
sum=30
deepasrir7@cloudshell:~/cloudlab$
```

CLOUD SHELL
Terminal cloudshell X +

GNU nano 7.2 app.py *
from http.server import HTTPServer, BaseHTTPRequestHandler
class MyHandler(BaseHTTPRequestHandler):
 def do_GET(self):
 self.send_response(200)
 self.end_headers()
 self.wfile.write(b"Hello from Google Cloud!")
HTTPServer(("", 8080), MyHandler).serve_forever()

^G Help

^O Write Out

^W Where Is

^K Cut

^T Execute

^C Location

^X Exit

^R Read File

^N Replace

^U Paste

^J Justify

^_ Go To Line

← → ↻

8080-cs-0e5c2bd9-054

Hello from Google Cloud!

```
deepasrir7@cloudshell:~/cloudlab$ iscpu
-bash: iscpu: command not found
deepasrir7@cloudshell:~/cloudlab$
deepasrir7@cloudshell:~/cloudlab$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:         46 bits physical, 48 bits virtual
Byte Order:            Little Endian
CPU(s):                2
  On-line CPU(s) list: 0,1
Vendor ID:             GenuineIntel
Model name:            Intel(R) Xeon(R) CPU @ 2.20GHz
CPU family:            6
Model:                 79
Thread(s) per core:    2
Core(s) per socket:    1
Socket(s):             1
Stepping:              0
top - 08:08:21 up 34 min,  0 user,  load average: 0.02, 0.03, 0.01
Tasks: 24 total,   1 running, 23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.5 us,   0.3 sy,   0.0 ni, 99.0 id,   0.0 wa,   0.0 hi,   0.2 si,   0.0 st
MiB Mem :  7947.4 total,  6295.1 free,   610.8 used,  1280.4 buff/cache
MiB Swap:   0.0 total,   0.0 free,   0.0 used.  7336.6 avail Mem

   695 deepasr+  10 -10    7340    3792    3544 S    0.0   0.0   0:00.00 bash
   696 deepasr+  10 -10    7340    3644    3384 S    0.0   0.0   0:00.00 start-shell.sh
   697 deepasr+  10 -10    6876    3516    3216 S    0.0   0.0   0:00.00 tmux
   699 deepasr+  10 -10    7276    3036    2256 S    0.0   0.0   0:00.14 tmux
   700 deepasr+  10 -10    9312    6316    3932 S    0.0   0.1   0:00.10 bash
   896 deepasr+  10 -10   14252    6364    4708 S    0.0   0.1   0:00.00 sshd
   989 deepasr+  10 -10    7340    3736    3484 S    0.0   0.0   0:00.00 bash
deepasrir7@cloudshell:~/cloudlab$ 4488    3888 S    0.0   0.1   0:00.00 bash
deepasrir7@cloudshell:~/cloudlab$
deepasrir7@cloudshell:~/cloudlab$ python3 app.py
```

```
   696 deepasr+  10 -10    7340    3644    3384 S    0.0   0.0   0:00.00 start-shell.sh
   697 deepasr+  10 -10    6876    3516    3216 S    0.0   0.0   0:00.00 tmux
   699 deepasr+  10 -10    7276    3036    2256 S    0.0   0.0   0:00.14 tmux
   700 deepasr+  10 -10    9312    6316    3932 S    0.0   0.1   0:00.10 bash
   896 deepasr+  10 -10   14252    6364    4708 S    0.0   0.1   0:00.00 sshd
   989 deepasr+  10 -10    7340    3736    3484 S    0.0   0.0   0:00.00 bash
deepasrir7@cloudshell:~/cloudlab$ 4488    3888 S    0.0   0.1   0:00.00 bash
deepasrir7@cloudshell:~/cloudlab$
deepasrir7@cloudshell:~/cloudlab$ python3 app.py
```

```
127.0.0.1 - - [28/Aug/2026 08:28:35] "GET /?authuser=0 HTTP/1.1" 200 -
^CTraceback (most recent call last):
  File "/home/deepasrir7/cloudlab/app.py", line 7, in <module>
    HTTPServer(("", 8080), MyHandler).serve_forever()
  File "/usr/lib/python3.12/socketserver.py", line 235, in serve_forever
    ready = selector.select(poll_interval)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/selectors.py", line 415, in select
    fd_event_list = self._selector.poll(timeout)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
KeyboardInterrupt

deepasrir7@cloudshell:~/cloudlab$
```

Directory listing for /?authuser=0

- [student.txt](#)

Welcome to Google Cloud Storage Demo

```
896 deepasr+ 10 -10 14252 6364 4708 S 0.0 0.1 0:00.00 sshd
989 deepasr+ 10 -10 7340 3736 3484 S 0.0 0.0 0:00.00 bash
deepasrir7@cloudshell:~/cloudlab$ 4488 3888 S 0.0 0.1 0:00.00 bash
deepasrir7@cloudshell:~/cloudlab$
deepasrir7@cloudshell:~/cloudlab$ python3 app.py

127.0.0.1 - - [28/Aug/2026 08:28:35] "GET /?authuser=0 HTTP/1.1" 200 -
^CTraceback (most recent call last):
  File "/home/deepasrir7/cloudlab/app.py", line 7, in <module>
    HTTPServer(("", 8080), MyHandler).serve_forever()
  File "/usr/lib/python3.12/socketserver.py", line 235, in serve_forever
    ready = selector.select(poll_interval)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/usr/lib/python3.12/selectors.py", line 415, in select
    fd_event_list = self._selector.poll(timeout)
                    ^^^^^^^^^^^^^^^^^^^^^^^^^
KeyboardInterrupt

deepasrir7@cloudshell:~/cloudlab$ mkdir storage_demo
deepasrir7@cloudshell:~/cloudlab$ cd storage_demo
deepasrir7@cloudshell:~/cloudlab/storage_demo$ echo "Welcome to Google Cloud Storage Demo"> student.txt
deepasrir7@cloudshell:~/cloudlab/storage_demo$ cat student.txt
Welcome to Google Cloud Storage Demo
deepasrir7@cloudshell:~/cloudlab/storage_demo$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
127.0.0.1 - - [28/Aug/2026 08:35:21] "GET /?authuser=0 HTTP/1.1" 200 -
127.0.0.1 - - [28/Aug/2026 08:35:57] "GET /student.txt HTTP/1.1" 200 -
□
```

```
deepasrir7@cloudshell:~/cloudlab/storage_demo$ echo "Welcome to Google Cloud
deepasrir7@cloudshell:~/cloudlab/storage_demo$ cat student.txt
Welcome to Google Cloud Storage Demo
deepasrir7@cloudshell:~/cloudlab/storage_demo$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
127.0.0.1 - - [28/Aug/2026 08:35:21] "GET /?authuser=0 HTTP/1.1" 200 -
127.0.0.1 - - [28/Aug/2026 08:35:57] "GET /student.txt HTTP/1.1" 200 -
^C
Keyboard interrupt received, exiting.
deepasrir7@cloudshell:~/cloudlab/storage_demo$
```

```
deepasrir7@cloudshell:~/cloudlab/storage_demo$ echo "Welcome to Google Cloud Storage
deepasrir7@cloudshell:~/cloudlab/storage_demo$ cat student.txt
Welcome to Google Cloud Storage Demo
deepasrir7@cloudshell:~/cloudlab/storage_demo$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
127.0.0.1 - - [28/Aug/2026 08:35:21] "GET /?authuser=0 HTTP/1.1" 200 -
127.0.0.1 - - [28/Aug/2026 08:35:57] "GET /student.txt HTTP/1.1" 200 -
^C
Keyboard interrupt received, exiting.
deepasrir7@cloudshell:~/cloudlab/storage_demo$ sqlite3 student.db
SQLite version 3.45.1 2024-01-30 16:01:20
Enter ".help" for usage hints.
sqlite> CREATE TABLE student(id INTEGER,name TEXT);
sqlite> INSERT INTO student VALUES(1,'Anu');
sqlite> INSERT INTO student VALUES(2,'Priya');
sqlite> SELECT * FROM student;
1|Anu
2|Priya
sqlite>
```

```
^C
Keyboard interrupt received, exiting.
deepasrir7@cloudshell:~/cloudlab/storage_demo$ sqlite3 student.db
SQLite version 3.45.1 2024-01-30 16:01:20
Enter ".help" for usage hints.
sqlite> CREATE TABLE student(id INTEGER,name TEXT);
sqlite> INSERT INTO student VALUES(1,'Anu');
sqlite> INSERT INTO student VALUES(2,'Priya');
sqlite> SELECT * FROM student;
1|Anu
2|Priya
sqlite> .quit
deepasrir7@cloudshell:~/cloudlab/storage_demo$
```